Assignment 4
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```python
import os
from dotenv import load_dotenv

load_dotenv()  # reads from .env in the same folder

GROQ_API_KEY   = os.getenv("GROQ_API_KEY")
GOOGLE_API_KEY = os.getenv("GOOGLE_API_KEY")
TAVILY_API_KEY = os.getenv("TAVILY_API_KEY")

# Make keys available as env vars for libraries that read them directly
os.environ["GROQ_API_KEY"]   = GROQ_API_KEY or ""
os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY or ""

# DeepEval uses OpenAI-compatible API — point it to Groq
os.environ["OPENAI_API_KEY"]  = GROQ_API_KEY or ""
os.environ["OPENAI_API_BASE"] = "https://api.groq.com/openai/v1"

print(f"GROQ_API_KEY:   {'SET ✓' if GROQ_API_KEY else 'NOT SET ✗'}")
print(f"GOOGLE_API_KEY: {'SET ✓' if GOOGLE_API_KEY else 'NOT SET ✗'}")
print(f"TAVILY_API_KEY: {'SET ✓' if TAVILY_API_KEY else 'NOT SET ✗'}")
```

[2] Python

```
GROQ_API_KEY:   SET ✓
GOOGLE_API_KEY: SET ✓
TAVILY_API_KEY: SET ✓
```

[3]

```
Knowledge base loaded: 1052 words, 6886 characters.
```

```
%pip install langchain.text_splitter
```

[7] Python

```
Split into 29 chunks.
Building FAISS vector store (downloading embedding model if needed)...
C:\Users\mohammed shazi\AppData\Local\Temp\ipykernel_17532\3813773578.py:16: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecat
  embeddings = HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")

Vector store built with 29 vectors.
Sample retrieval test:
  Chunk 1: Chandrayaan-2 was launched on July 22, 2019. It consisted of an orbiter, a lander named Vikram, and a rover named Pragya...
  Chunk 2: Chandrayaan-1, launched on October 22, 2008, was India's first lunar mission. It carried 11 payloads, including instrume...
  Chunk 3: Chandrayaan-3 was launched on July 14, 2023, aboard a LVM3 rocket. On August 23, 2023, the Vikram lander successfully so...
  Chunk 4: . The Pragyan rover deployed from the lander and confirmed the presence of sulphur and other elements in the lunar soil....
```

```python
import json
from crewai import Agent, Task, Crew, LLM
from crewai.tools import tool
from langchain_groq import ChatGroq

# ── LLM Configuration ──────────────────────────────────────
# Using llama-3.3-70b-versatile for best quality; 8b-instant as a backup
crew_llm = LLM(
    model="groq/llama-3.3-70b-versatile",
    temperature=0.2,
    max_tokens=1024,
    api_key=GROQ_API_KEY
)

print("CrewAI LLM configured: groq/llama-3.3-70b-versatile")
```

Python

```
CrewAI LLM configured: groq/llama-3.3-70b-versatile
```

```
RAG search tool defined.

Test query: 'What was India's first lunar mission?'
Answer: Chandrayaan-1, launched on October 22, 2008, was India's first lunar mission.
Retrieved 4 context chunks.
```

```
Testing RAG Agent on 3 sample questions...

Tool isro_knowledge_base_search executed with result: {"answer": "ISRO was established on August 15, 1969, under the leadership of Vikram Sarabhai, who

Q: When was ISRO founded and who founded it?

A: ISRO was established on August 15, 1969, under the leadership of Vikram Sarabhai, who is known as the father of the Indian space programme.
   [4 context chunks retrieved]

Tool isro_knowledge_base_search executed with result: {"answer": "The total cost of Chandrayaan-3 was approximately \u20b9615 crore (around $75 million
Maximum iterations reached. Requesting final answer.

Q: What was the cost of the Chandrayaan-3 mission?
A: The total cost of Chandrayaan-3 was approximately ₹615 crore (around $75 million USD).
   [4 context chunks retrieved]

Maximum iterations reached. Requesting final answer.

Q: What record did PSLV-C37 set in February 2017?
A: PSLV-C37 set a record by launching 104 satellites in a single mission in February 2017.
   [4 context chunks retrieved]
```

```
######################################################################
Running Q1: What was the significance of Chandrayaan-3's landing site and what did it discover?
######################################################################

======================================================================
QUESTION: What was the significance of Chandrayaan-3's landing site and what did it discover?
======================================================================

[AGENT 1] RAG Retriever...
  Answer: Chandrayaan-3's landing site was significant because it was near the Moon's south pole, making India the first and only country to land near
  Retrieved 4 context chunks.
```

```
Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"eb6c01b62f1341478256e0b25dbaa475","version_major":2,"version_minor":0}

[AGENT 2] Quality Evaluator...

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"fb38631729244745ab836e4b885a5f2a","version_major":2,"version_minor":0}

  Faithfulness : 0.667 (FAIL)
  Relevancy    : 1.0 (PASS)
  Verdict      : FAIL
  Failures     : ["Faithfulness FAILED (score=0.667): The score is 0.67 because the actual output incorrectly claimed Chandrayaan-3's landing site made

[AGENT 3] Revisor (activating due to FAIL)...

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"961644052b5c4a3e9b296455197f5bef","version_major":2,"version_minor":0}

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"9cb5fc441d894bfe826daa9b603ac1f5","version_major":2,"version_minor":0}

  Revised Answer: The significance of Chandrayaan-3's landing site lies in its proximity to the Moon's south pole. According to the context, India is t
  After Revision — Faithfulness: 0.6, Relevancy: 0.667
  Final Verdict: FAIL
```

```
######################################################################
Running Q2: What is NavIC and when did it become fully operational?
######################################################################

======================================================================
QUESTION: What is NavIC and when did it become fully operational?
======================================================================

[AGENT 1] RAG Retriever...
  Answer: NavIC (Navigation with Indian Constellation) is India's own regional satellite navigation system, similar to GPS. It became fully operational
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"b31db32a1edf40d8964ae59677115b49","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"f71b72d48eb7427584be7cf245b82260","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 1.0 (PASS)
  Verdict      : PASS

[VERDICT] PASS — no revision needed.
```

```
######################################################################
Running Q3: How does the GSLV Mark III (LVM3) compare to PSLV in payload capacity?
######################################################################

======================================================================
QUESTION: How does the GSLV Mark III (LVM3) compare to PSLV in payload capacity?
======================================================================

[AGENT 1] RAG Retriever...
  Answer: The GSLV Mark III (LVM3) has a higher payload capacity compared to the PSLV. The LVM3 can place up to 4,000 kg in geostationary transfer orbi
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"64da580f535a4a04b814b580fd75853f","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"e7958ea06f2049049cc874dbc485261b","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 1.0 (PASS)
  Verdict      : PASS

[VERDICT] PASS — no revision needed.
```

```
######################################################################
Running Q4: What is Gaganyaan and what was the TV-D1 test?
######################################################################

======================================================================
QUESTION: What is Gaganyaan and what was the TV-D1 test?
======================================================================

[AGENT 1] RAG Retriever...
  Answer: Gaganyaan is India's human spaceflight programme, which aims to send Indian astronauts (called Gaganauts) to low Earth orbit. The TV-D1 test,
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"1c8026a97ca24c53833bc1de57738407","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"d2cbbbd71fe945718ee4201069a61db4","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 1.0 (PASS)
  Verdict      : PASS

[VERDICT] PASS — no revision needed.
```

```
######################################################################
Running Q5: What was Mangalyaan's total mission cost and why was it significant?
######################################################################

====================================================================
QUESTION: What was Mangalyaan's total mission cost and why was it significant?
====================================================================

[AGENT 1] RAG Retriever...
  Answer: Mangalyaan's total mission cost was approximately ₹450 crore ($74 million USD), making it significant as the least expensive Mars mission in
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"c366f460c470470e893959edc958db58","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"b13a4698809c4d7da075fdf94adbe7f6","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 1.0 (PASS)
  Verdict      : PASS

[VERDICT] PASS — no revision needed.
```

```
######################################################################
Running ADVERSARIAL: What is NASA's current budget for the Artemis programme in 2024?
######################################################################

====================================================================
QUESTION: What is NASA's current budget for the Artemis programme in 2024?
====================================================================

[AGENT 1] RAG Retriever...
  Answer: I don't have sufficient information in my knowledge base to answer this question.
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"131b5504f1d9476abf038622de79e5bf","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"ddc696cd193649a992aec72f7c8d6ea5","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 0.0 (FAIL)
  Verdict      : FAIL
  Failures     : ['Answer Relevancy FAILED (score=0.0): The score is 0.00 because the actual output explicitly stated a lack of information to address

[AGENT 3] Revisor (activating due to FAIL)...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"af1c4a2113df43b4b82301ba3bbbb1d9","version_major":2,"version_minor":0}

```
  Revised Answer: The retrieved context does not provide specific information about NASA's current budget for the Artemis programme in 2024. However, i
  After Revision — Faithfulness: 0.8, Relevancy: 0.833
  Final Verdict: PASS

######################################################################
Running ADVERSARIAL: Who is the current Director General of ESA?
######################################################################

======================================================================
QUESTION: Who is the current Director General of ESA?
======================================================================

[AGENT 1] RAG Retriever...
  Answer: I don't have sufficient information in my knowledge base to answer this question.
  Retrieved 4 context chunks.
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"9f6574b8b2ef4d8d9fb983d0af2bafce","version_major":2,"version_minor":0}

```
[AGENT 2] Quality Evaluator...
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"1dffa9155d9b483e9d7336725ea58c48","version_major":2,"version_minor":0}

```
  Faithfulness : 1.0 (PASS)
  Relevancy    : 0.0 (FAIL)
  Verdict      : FAIL
  Failures     : ['Answer Relevancy FAILED (score=0.0): The score is 0.00 because the actual output explicitly stated a lack of information, making it
```

Generate | Code | Markdown | Run All | Outline ··· Select Kernel

```
···   Faithfulness : 1.0 (PASS)
      Relevancy    : 0.0 (FAIL)
      Verdict      : FAIL
      Failures     : ['Answer Relevancy FAILED (score=0.0): The score is 0.00 because the actual output explicitly stated a lack of information, making it

    [AGENT 3] Revisor (activating due to FAIL)...
```

··· Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"4464a00243634878bae109432c91d78e","version_major":2,"version_minor":0}

··· Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"c283bc7923704fae80530cfe5dc2247e","version_major":2,"version_minor":0}

```
···   Revised Answer: The provided Retrieved Context does not contain information about the European Space Agency (ESA) or its current Director General. Th
      After Revision — Faithfulness: 1.0, Relevancy: 0.5
      Final Verdict: FAIL

    ======================================================================
    All 7 questions processed.
```

```
ADVERSARIAL QUESTION ANALYSIS
======================================================================
These questions have answers that are NOT in the knowledge base.
Expected behavior: system should acknowledge lack of information.

QUESTION: What is NASA's current budget for the Artemis programme in 2024?

FINAL ANSWER:
The retrieved context does not provide specific information about NASA's current budget for the Artemis programme in 2024. However, it does mention tha

SCORES: Faithfulness=0.8, Relevancy=0.833, Verdict=PASS
Graceful handling: NO — may have hallucinated ✗
----------------------------------------------------------------------
QUESTION: Who is the current Director General of ESA?

FINAL ANSWER:
The provided Retrieved Context does not contain information about the European Space Agency (ESA) or its current Director General. The context primaril

SCORES: Faithfulness=1.0, Relevancy=0.5, Verdict=FAIL
Graceful handling: NO — may have hallucinated ✗
----------------------------------------------------------------------
```